Cross-Resolution Change Detection in Remote Sensing via Unequal Relationships From a Frequency Perspective

Lichen Ning, Qing Zhou, Qi Wang, *Senior Member, IEEE*, Junyu Gao, *Member, IEEE*, and Xuelong Li, *Fellow, IEEE*

***Abstract*— Cross-resolution change detection (CRCD) identifies changes between bitemporal images with different resolutions, which provide better adaptability to real-world applications than conventional change detection (CD). Existing CRCD methods first align the resolution of different temporal images, then employ the Siamese network. Experiments conducted from a frequency-domain perspective validate that resize operations disrupt the data distribution and result in performance degradation of the Siamese network. Further experiments reveal the resolution-invariant temporal and spatial unequal relationship between bitemporal images. Specifically, spatial specificity information within a specific temporal domain is more critical for CRCD, i.e., high-frequency components in a specific temporal domain are closely related to change label. And this unequal relationship exhibits invariance in resolution. On this basis, we propose the Fourier and wavelet transform-based inequality Siamese network (FWISN) to address the performance degradation observed in Siamese networks on CRCD, leveraging the inequality between bitemporal images to improve network performance. FWISN includes a frequency reconstruction (FRC) stage, in which high-frequency components of a given temporal-domain image are extracted and reconstructed using our proposed high-frequency attention (HFA) module. We further propose the wavelet transform-based frequency learning block (WFB), which enhances high-frequency features and is integrated into both the encoder (WFB-E) and the decoder (WFB-D) of the network. The experiments demonstrate state-of-the-art performance, compared with methods specifically designed for cross-resolution tasks, FWISN achieving $F1$/intersection over union (IoU) improvements of 2.15/3.51, 2.50/4.59, and 1.93/1.73 on the LEVIR-CD (4×), SV-CD (8×), and DE-CD (3.3×) tests, respectively. Furthermore, in the continuous CRCD task, FWISN achieves $F1$/IoU improvements of 8.39/11.24 on the LEVIR-CD (8×) test. Our code will be public on https://github.com/blacksheep182/NN_FWISN**

***Index Terms*— Cross-resolution change detection (CRCD), frequency-domain learning, remote sensing.**

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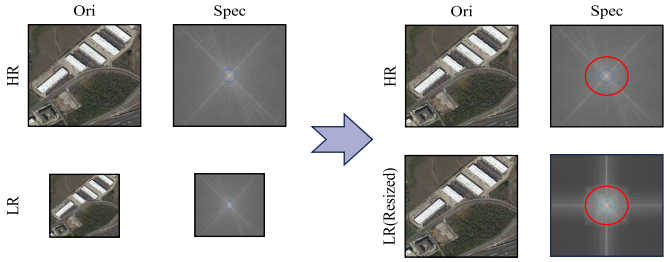
I. Introduction

CHANGE detection (CD) involves identifying regions within multitemporal remote sensing image (RSI) pairs and categorizing them into binary labels at the pixel level: changed or unchanged [1]. CD plays a crucial role in various applications, including urban development planning, environmental surveying, and disaster monitoring. Due to its significance, CD has garnered sustained attention from both academia and industry.

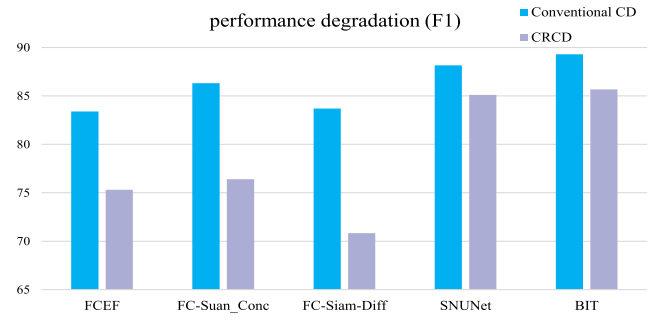
However, conventional CD methods are often customized for RSI pairs with the same resolution, which may not meet the requirements of real-world applications. For example: 1) RSIs from different sensors typically have varying resolutions; 2) earlier sensors typically have lower spatial resolution due to technological limitations; hence, for some long-term CD tasks (e.g., geomorphological changes), it is necessary to process cross-resolution images (LR pre-event RSI and HR postevent RSI); and 3) due to the limited spatial coverage and poor real-time performance of HR images, certain CD tasks that require rapid response (e.g., disaster detection) often require processing cross-resolution images (HR pre-event RSI and LR postevent RSI). Therefore, cross-resolution change detection (CRCD), which identifies regions of change in images of varying resolutions, offers better adaptability to real-world applications than conventional CD.

Given the superior performance of the Siamese network architecture in conventional CD, most of the existing methods for CRCD seek to adapt this framework. However, experimental comparisons show significant performance degradation when the conventional Siamese CD network is applied directly to CRCD, as illustrated in Fig. 1(b). This observation suggests that the Siamese network architecture may not be ideally suited for CRCD tasks. The inherent need for consistent input in Siamese networks forces some methods to standardize image resolutions by resizing images of varying resolutions to a common size, either by downsampling HR images [2], [3], or by upsampling LR images [4], or applying learnable approaches [5], [6]. However, these operations may distort the original data distribution, thus compromising input consistency for the Siamese network. Some methods address resolution discrepancies by mapping multitemporal-domain images into

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(a) LR and HR refer to low-resolution and high-resolution images, respectively, while Ori and Spec denote the original image and spectrogram.



(b) Comparison of the performance of several Siamese networks on conventional CD and CRCD. Conventional CD is evaluated on LEVIR-CD, while CRCD is tested on LEVIR-CD at 4× resolution.

Fig. 1. (a) Resize operation disrupts the frequency distribution, which subsequently leads to (b) degradation in $F1$-score performance on LEVIR-CD.

a shared feature space [7], [8], [9]. However, these methods still fail to adequately capture resolution-invariant features.

In this context, recent advances in tiny object detection provide valuable insight into handling the challenges of resolution-invariant features. For example, the proposed tiny target detection method [10], [11] for dealing with low-resolution targets and weakly characterized targets is compatible with the need to identify change regions with fuzzy edges, small sizes, and weak differences in CRCD.

To mitigate the performance degradation problem inherent in current methods for CRCD tasks, we aim to propose a novel CD framework explicitly designed to adapt to different spatial resolutions. This framework seeks to address the input images with different resolutions by leveraging advanced resolution normalization techniques, ensuring precise alignment and effective utilization of features extracted from images with diverse spatial scales. By addressing the challenges posed by resolution discrepancies, the proposed framework aims to maintain robust detection capabilities across diverse resolution scenarios.

Frequency-domain learning methods offer the advantage of global modeling [12], [13]. This capability of global modeling helps bridge the gap in the contextual information of LR images, which is a significant benefit in super-resolution tasks. Given the potential of frequency-domain learning methods for processing LR images, we investigate the application of frequency learning to CRCD. By analyzing the spectrograms of images with varying resolutions, we observe the changes in data distribution through their frequency-domain representations. As illustrated in Fig. 1(a), despite resolution differences, the images show similar frequency distributions. However, when the LR image is resized to match the HR image, the spectrogram of the resized LR image lacks high-frequency components compared with the HR one. This indicates that the resizing process disrupts the frequency distribution of the LR image, which in turn impacts the overall data distribution. As a result, this alteration undermines the effectiveness of the Siamese network structure in CRCD.

In addition, we apply high-pass filtering to extract high-frequency components of bitemporal images, as shown in Fig. 2. The results clearly demonstrate that high-frequency components of specific time-domain images align closely with the change labels. This correspondence is also observable in the spectrograms, where these images show phase distributions that are more similar to the labels. More importantly, this correspondence exhibits invariance in resolution: Despite the resized LR image exhibiting weaker high-frequency components, it continues to align well with the change labels. This observation leads us to derive a temporal and spatial unequal relationship in CRCD from the frequency-domain perspective: *spatially specific information in certain temporal domains plays a more significant role in CRCD*. It is noteworthy that, despite deriving this relationship from the frequency domain, we refer to it as spatially unequal because high-frequency components correspond to edge information in the image. This edge information is usually aligned with semantic features such as buildings, vegetation, and others, essentially reflecting a spatial-domain feature.

Our hypothesis is that the resolution-invariant temporal and spatial unequal relationship can mitigate resolution discrepancies to some extent. Consequently, we aim to design the CRCD framework around this unequal relationship to ensure robust detection performance across different resolutions. Therefore, we propose the Fourier and wavelet transform-based inequality Siamese network (FWISN). Specifically, we introduce a frequency reconstruction (FRC) stage, which reconstructs the image by utilizing an unequal relationship to emphasize semantic information that is invariant to resolution. In the FRC, we propose the high-frequency attention (HFA) mechanism, which utilizes high-frequency information to generate the attention map to reconstruct high-frequency features. The high-frequency component of the attention map is derived from an image enriched with high-frequency components. In FRC, bitemporal images are jointly modeled with high-frequency components from a specific spatial and temporal domain, generating resolution-invariant semantic features. In addition, we propose the wavelet transform-based frequency learning block (WFB) and its specialized versions, WFB-E and WFB-D, which further enhance the model's adaptation to CRCD. WFB-E is designed for the encoder, where interactions between bitemporal features are omitted. This enables independent high-frequency feature extraction. Both WFB-E and WFB-D ensure comprehensive fusion and accurate spatial information exchange within change regions, ultimately enhancing the model's CD capabilities across varying resolutions. Our contribution can be summarized as follows.

1) From a frequency-domain perspective, we observe temporal and spatial unequal relationships in CRCD: spatially specific semantic information in a particular temporal domain plays a more significant role in CRCD.
2) We propose FWISN, which introduces a frequency-domain learning into CRCD and addresses the performance degradation in Siamese networks.

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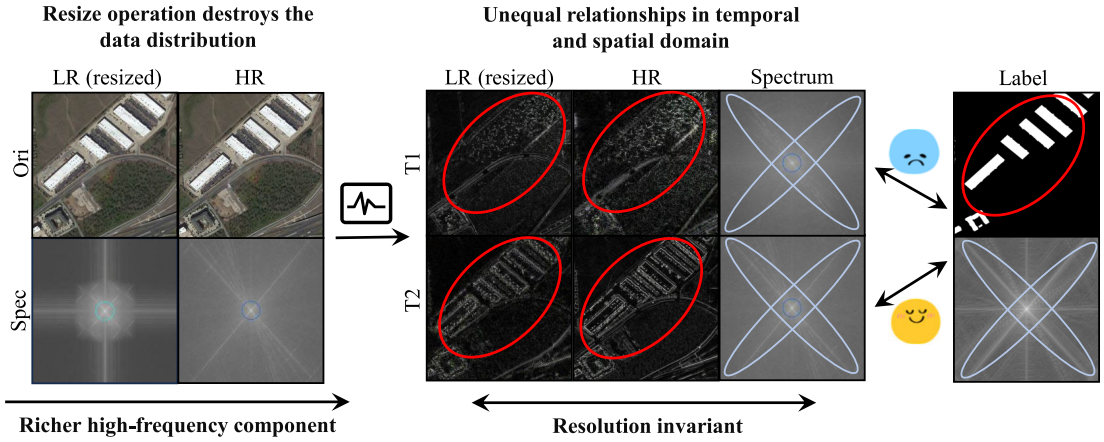


Fig. 2. Resize operation destroys the data distribution, but spatial specificity information within a specific temporal-domain match better with label and is resolution-invariant, which is critical in CRCD task.

3) We propose FRC, based on HFA, to reconstruct images. We also proposed WFB to extract high-frequency features, which helps in extracting resolution-invariant features by leveraging unequal relationships in CRCD.
4) Experiments are conducted on three CD datasets to validate the efficacy of our proposed method, achieving superior performance under cross-resolution scenarios, particularly excelling in continuous cross-resolution scenarios.

II. Related Works

A. Siamese Network in CD

Deep learning has achieved remarkable advances in the field of remote sensing [14], [15], [16]. Many classical deep learning methods, such as convolutional neural networks (CNNs) [17], fully convolutional networks (FCNs) [18], and transformers [19], have been extensively employed in CD tasks [20]. Initially, CD is framed as a segmentation problem, leading to the application of classical segmentation algorithms such as U-Net [8]. To address the bitemporal characteristics inherent in CD, Siamese network architectures have been proposed for processing pairs of images [21], [22]. Although single-stream networks have shown effectiveness in other remote sensing applications [23], the dual-stream Siamese architecture remains the predominant approach for CD.

Recent investigations in CD have focused on enhancing the contextual modeling capabilities of networks. One strategy involves the utilization of multiscale dilated convolutions [24], [25], which expands the receptive field of CNNs, thereby enabling the acquisition of more comprehensive contextual information. In addition, some studies implement attention mechanisms in CD tasks to strengthen the network's capacity for processing spatial–temporal data [26], [27], [28]. Given the superior global modeling capabilities of transformers, the bitemporal image transformer (BIT) [29] employs a transformer network to learn semantic features in sparse tokens extracted from bitemporal RSI. Furthermore, Changeformer [30] proposes a purely transformer-based CD network, underscoring the importance of contextual modeling for effective CD. In addition, some works [31], [32], [33], [34] have explored the application of twin networks in few-shot/zero-shot tasks.

However, most existing methodologies are designed for fixed resolution scenarios with fixed resolution differences, which makes them inadequate for scenarios with varying resolution discrepancies. In practical applications, CRCD has garnered considerable attention. Most current methods still rely on Siamese architectures, although resolution matching is conducted during the image input phase (e.g., HR to LR [2], LR to HR [4], and super-resolution techniques [5], [6]). Some studies have explored the alignment of feature space for CRCD, such as using transformers to align semantic features by combining upsampled low-resolution and high-resolution features [7], and leveraging implicit neural representation to achieve stable multiscale feature learning [35].

Nonetheless, these methods do not adequately address resolution-invariant information, leading to performance degradation in Siamese networks. Our approach remedies this limitation by focusing on the extraction of resolution-invariant high-frequency features.

B. Resolution-Invariant Frequency-Domain Learning

The Fourier transform has long served as a foundational tool in digital image processing. Lots of studies have investigated its integration into model architectures in many computer vision tasks. The FFC [36] architecture introduces a local Fourier unit which performs the frequency convolution method into CNNs. This method effectively addresses with varying scales, while strengthening the model's capacity to capture both large and small-scale features in a complementary manner. GFNet [37] proposes the global filter network, which leverages learnable global filters to perform elementwise multiplication with features in the frequency domain. SpectFormer [38] integrates spectral components with multihead self-attention within the original ViT [39] architecture, thereby augmenting its representational capabilities.

Frequency-domain learning techniques have also been applied to super-resolution tasks. FNNSR [40] proposes a full design of frequency-domain-based neural network. Subsequently, IFNNSR [41] enhances FNNSR by partitioning

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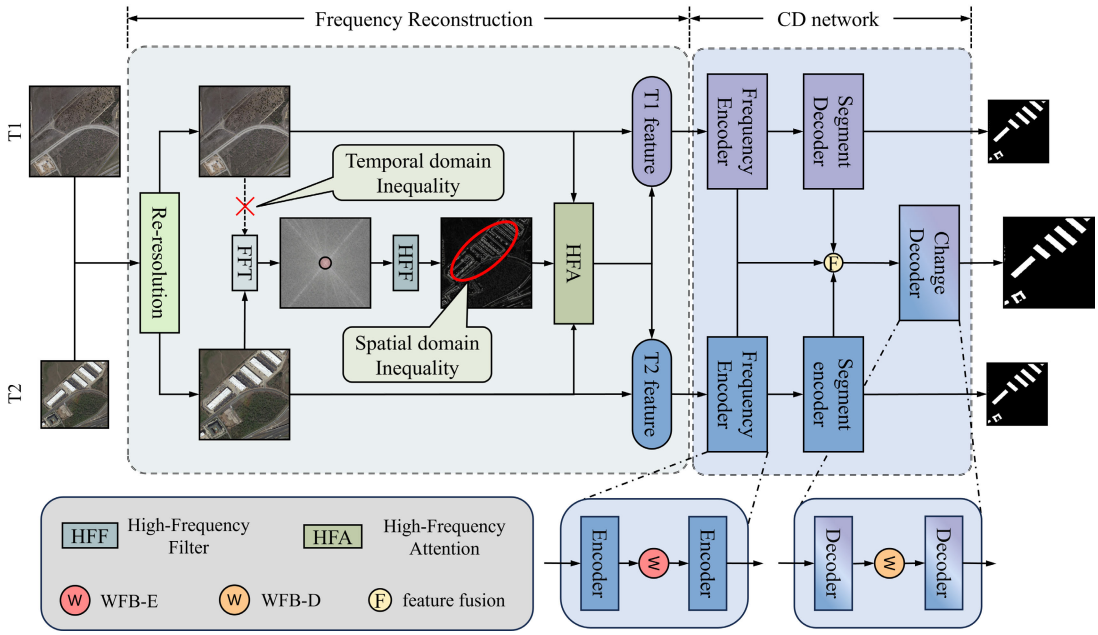


Fig. 3. Overview of the FWISN. It comprises two main parts: frequency-domain filtering-based image reconstructed block and CD network with frequency-domain convolution-based encoder and decoder.

its weight matrices into four quadrants. This method shares parameters, reducing the total number of parameters and improving computational efficiency. Inspired by FFC, SwinFIR [42] integrates fast Fourier transform (FFT) capabilities with spatial transformations. It employs a complementary dual-branch structure embedded within the SwinTR framework [43], yielding impressive results. These methodologies effectively demonstrate the potential of frequency-domain learning for processing images across diverse resolutions.

Some studies have employed frequency-domain learning methods in remote sensing [44], [45], [46], [47]. These methods typically leverage purely frequency-domain-based representations. However, they often overlook the roles of different frequency components in the context of RSI. For example, CNNs exhibit heightened sensitivity to low-frequency components of an image [48], which contrasts with intuitive expectations. Our method incorporates a convolution module designed to amplify the high-frequency components of RSI, enabling the network to focus more effectively on this critical information in the images.

III. Methodology

A. Research Objective

CRCD is a specialized task in RSI analysis, designed to address the challenge of identifying changes between bitemporal images with different spatial resolutions. Unlike conventional CD, which typically assumes that both temporal images have the same spatial resolution, CRCD is better aligned with the constraints and demands of real-world applications, where images are captured by different sensors or where temporal and technological constraints lead to resolution mismatches. The goal of CRCD is to effectively bridge the resolution gap and accurately detect change regions.

Formally, given bitemporal RSIs T_1 and T_2, with different spatial resolutions $H_1 \times W_1$ and $H_2 \times W_2$, and a consistent number of spectral or channel dimensions C, the inputs can be represented as: $T_1 \in \mathbb{R}^{H_1 \times W_1 \times C}$ and $T_2 \in \mathbb{R}^{H_2 \times W_2 \times C}$. The objective of the model is to predict a unified segmentation map S' at a target resolution $H \times W$, which closely approximates the ground-truth segmentation map S, where $S' \in \mathbb{R}^{H \times W}$ and $S \in \mathbb{R}^{H \times W}$. Each pixel in the predicted segmentation map S' is assigned to one of the predefined binary classes $Y = \{y_0, y_1\}$, where y_0 represents the "unchanged" class and y_1 represents the "changed" class. Thus, S' constitutes a binary segmentation map with

$$S'_{i,j} \in \{y_0, y_1\} \quad \forall (i, j) \in [1, H] \times [1, W].$$

In the context of continuous CRCD, the task further introduces a domain adaptation challenge, wherein the model is trained using inputs at one pair of spatial resolutions and evaluated on inputs at different resolutions. Specifically, during training, the model processes inputs: $T_1 \in \mathbb{R}^{H_1 \times W_1 \times C}$ and $T_2 \in \mathbb{R}^{H_1 \times W \times C}$ while during testing, it encounters inputs: $T'_1 \in \mathbb{R}^{H'_1 \times W'_1 \times C}$ and $T'_2 \in \mathbb{R}^{H'_2 \times W'_2 \times C}$ The output is a unified segmentation map $S \in \mathbb{R}^{H \times W}$. This setup demands that the model exhibits robust resolution-invariant feature learning capabilities, enabling it to generalize effectively across varying spatial resolutions.

B. Overall Model Architecture

The overview of the FWISN is illustrated in Fig. 3. It consists of two main components: FRC stage and CD network.

In the FRC stage, in order to guarantee input consistency for the network, we first resize LR image to match HR ones through the use of interpolation. In order to highlight the high-frequency component (corresponding to change information) of the image in the stage of image reconstruction,

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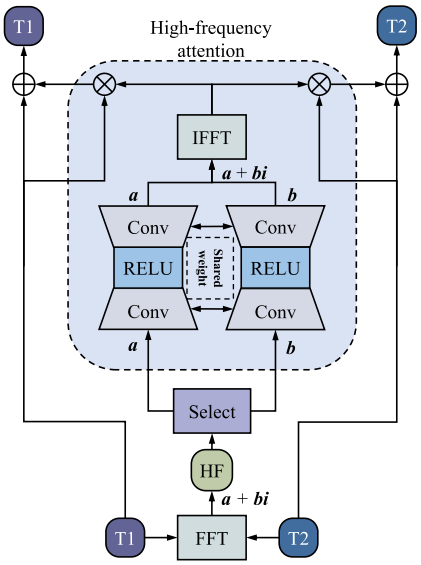


Fig. 4. Overview of the HFA. The bitemporal HF component comes from the branch with richer HF components. Finally, we get the reconstructed image highlighting the high-frequency components. The standard convolution modules are not designed to operate on complex-valued inputs, we apply a shared-weight convolution module separately to the real and imaginary components during the FFT process.

we filter the high-frequency component of bitemporal RSI pairs. We then propose a frequency-domain select mechanism to select the branch with more abundant high-frequency components to construct the attention map of our proposed HFA, so as to highlight the high-frequency information of the particular temporal-domain RSI and to leverage the unequal relationship between the bitemporal RSI.

In the CD network, we adopt the overall architecture of the exchanging dual encoder–decoder [28], which consists of a dual encoder–decoder and a change decoder. We integrate our proposed WFB into both the dual encoder (WFB-E) and the change decoder (WFB-D). The primary difference between WFB-E and WFB-D lies in the exchange of high-frequency features processed by the wavelet transform (WT), which facilitates improved feature fusion and enables the network to capture spatial unequal relationships between bitemporal features. To extract multiscale features with rich local information, we utilize a WT-based frequency-domain learning method instead of FFT. However, as the encoder layers deepen, the feature maps become smaller, reducing the effectiveness of WT. Therefore, we apply the frequency-domain learning method only in the initial layers of the encoder. For the deeper encoder layers, we employ a channel exchange module to enhance the interaction between bitemporal features.

C. Frequency-Domain-Based Image Reconstruction

We propose FRC based on HFA, as shown in Fig. 4.

In FRC, we first apply the FFT on the bitemporal image and then set a specific frequency threshold to isolate the HF component. Finally, we get the high-frequency map $\mathbf{X_{hf}}$. The $\mathbf{X_{hf}}$ can be formulated as

$$z_f = f(x_f, y_f) = \text{FFTShift}(\text{FFT}(x, y)) \tag{1}$$

$$\mathbf{X_{hf}} = \{z_h = f(x_h, y_h) | (x_h, y_h) \in (x_f, y_f), \quad (x_h - w/2)^2 + (y_h - h/2)^2 < R^2\} \tag{2}$$

where (x_h, y_h) and (x_f, y_f) denote the coordinates in the HF feature map and frequency map. The parameters w and h represent the width and height of the feature map and frequency map, while R represents the frequency threshold of the filter, which can be formulated as follows:

$$R = (w + h)/\theta \tag{3}$$

where w and h represent width and height of the input image and θ is the parameter that controls the high-frequency component.

Then we introduce HFA to generate the attention map from HF components. The HF features extracted by HFA are first combined with the original image through elementwise multiplication and then added back to original images. This process ultimately yields a reconstructed image with enhanced high-frequency components, which can be formulated as follows:

$$\mathbf{M_{HFA}} = \text{Conv}(\text{ReLU}(\text{Conv}(\mathbf{X_{hf}}))) \tag{4}$$

$$\mathbf{F_{attn1}}, \mathbf{F_{attn2}} = \mathbf{M_{HFA}} \otimes \mathbf{x_1}, \mathbf{M_{HFA}} \otimes \mathbf{x_2} \tag{5}$$

$$\mathbf{F_{RC1}}, \mathbf{F_{RC2}} = \mathbf{F_{attn1}} + \mathbf{x_1}, \mathbf{F_{attn2}} + \mathbf{x_2} \tag{6}$$

where $\mathbf{M_{HFA}}$ denotes the attention map generated by the HFA, $\mathbf{x_1}, \mathbf{x_2}$ represent the T1 and T2 images, and $\mathbf{F_{RC1}}$ and $\mathbf{F_{RC2}}$ denote the reconstructed features of the bitemporal images. Note that the bitemporal HF components come from the branch with richer HF components. To quantitatively evaluate the distribution characteristics of high-frequency information, we compute the average channelwise entropy for the bitemporal RSI. We first normalize them using the softmax function along the channel dimension

$$\tilde{A}_{c,i,j} = \frac{\exp(A_{c,i,j})}{\sum_{k=1}^{C} \exp(A_{k,i,j})}, \quad \tilde{B}_{c,i,j} = \frac{\exp(B_{c,i,j})}{\sum_{k=1}^{C} \exp(B_{k,i,j})} \tag{7}$$

where $A_{c,i,j}$ and $B_{c,i,j}$ represent the value of $\mathbf{X_{hf}}$ extracted from RSI T1 and T2 at each spatial location (i, j), c represents channel, and $\tilde{A}$ and $\tilde{B}$ represent the channelwise distributions for sample. Next, the Shannon entropy is computed at each spatial location and then averaged over the entire spatial domain

$$H_A = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left(- \sum_{c=1}^{C} \tilde{A}_{c,i,j} \log \tilde{A}_{c,i,j} \right) \tag{8}$$

$$H_B = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} \left(- \sum_{c=1}^{C} \tilde{B}_{c,i,j} \log \tilde{B}_{c,i,j} \right). \tag{9}$$

Finally, we compute the entropy difference between the two tensors for each sample as

$$\Delta H = H_A - H_B. \tag{10}$$

A positive value ΔH indicates that high-frequency components in B are more concentrated, and we utilize the T2 image to compute $\mathbf{X_{hf}}$. While a negative value implies that A has a more focused representation, and we employ the T1 image to compute $\mathbf{X_{hf}}$. This entropy-based metric allows us to quantitatively assess which representation exhibits stronger

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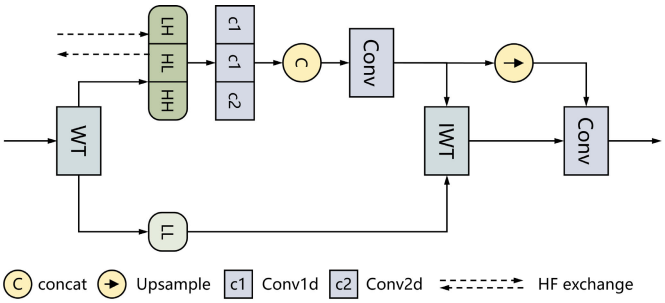


Fig. 5. Structure of WFB. WFB is applied to the encoder and change decoder. When WFB is used in the decoder, the high-frequency components are exchanged between bitemporal branches.

channelwise high-frequency components, potentially reflecting sharper semantic edge focus.

In the FRC process, we establish temporal and spatial inequality relationships between bitemporal images. Given that $\mathbf{X_{hf}}$ in (4) is used to construct the attention map and originates from a specific temporal branch, this results in a temporal inequality between bitemporal images during the FRC process. Simultaneously, we focus on reconstructing only the high-frequency components of the image, which play a crucial role in capturing spatial inequalities.

D. WT-Based Frequency Learning Block

Fig. 5 illustrates the structure of WFB. WFB is applied to WFB-E and WFB-D. The interactions between bitemporal features are omitted in WFB-E. While high-frequency components are exchanged in WFB-D to achieve better feature fusion. Given the enhanced local modeling capacity and improved computational efficiency, the WT is considered as a superior alternative to FFT. In this block, we utilize the Haar WT, which exhibits both efficiency and simplicity [13], [49], [50]. The formulas for Haar WT filters are

$$\mathbf{f_{LL}} = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}, \quad \mathbf{f_{LH}} = \frac{1}{2}\begin{bmatrix}1 & -1\\ 1 & -1\end{bmatrix}$$
$$\mathbf{f_{HL}} = \frac{1}{2}\begin{bmatrix}1 & 1\\ -1 & -1\end{bmatrix}, \quad \mathbf{f_{HH}} = \frac{1}{2}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix} \tag{11}$$

where $\mathbf{f_{LL}}$ serves as a low-pass filter, while $\mathbf{f_{LH}}$, $\mathbf{f_{HL}}$, and $\mathbf{f_{HH}}$ serve as a collection of high-pass filters, while L/H denotes the horizontal or vertical low/high pass filters. To establish spatial inequality, we perform feature extraction only on the high-frequency components with convolution to obtain $\mathbf{hf_{LH}}$, $\mathbf{hf_{HL}}$, and $\mathbf{hf_{HH}}$, leaving the low-frequency components $\mathbf{f_{LL}}$ unchanged. This process can be expressed in terms of the following equations:

$$\mathbf{f_{LL}}, \mathbf{f_{LH}}, \mathbf{f_{HL}}, \mathbf{f_{HH}} = \mathrm{WT}(\mathbf{x_{in}}) \tag{12}$$
$$\mathbf{hf_{LH}} = \mathrm{Conv1d}(\mathbf{f_{LH}}) \tag{13}$$
$$\mathbf{hf_{HL}} = \mathrm{Conv1d}(\mathbf{f_{HL}}) \tag{14}$$
$$\mathbf{hf_{HH}} = \mathrm{Conv}(\mathbf{f_{HH}}). \tag{15}$$

In addition, these high-frequency components are concatenated and convolved to create a consistent set of high-frequency features. Then, we perform the inverse WT (IWT) to reconstruct the features

$$\mathbf{hf} = \mathbf{hf1_{LH}}, \mathbf{hf1_{HL}}, \mathbf{hf1_{HH}}$$
$$= \mathrm{Conv}(\mathrm{Concat}(\mathbf{hf_{LH}}, \mathbf{hf_{HL}}, \mathbf{hf_{HH}})) \tag{16}$$
$$\mathbf{f_{rc}} = \mathrm{IWT}(\mathbf{f_{LL}}, \mathbf{hf1_{LH}}, \mathbf{hf1_{HL}}, \mathbf{hf1_{HH}}). \tag{17}$$

Finally, these high-frequency features are upsampled and then concatenated with the reconstructed features. The concatenated result is further processed through a convolution operation to generate the final output

$$\mathbf{x_{out}} = \mathrm{Conv}(\mathrm{Concat}(\mathbf{f_{rc}}, \mathbf{hf})). \tag{18}$$

In the WFB-D module, to better integrate the features of the bitemporal branches, we exchange their high-frequency components. This exchange enhances the model's ability to capture complementary temporal information, and (17) becomes

$$\mathbf{f_{rc1}} = \mathrm{IWT}(\mathbf{f_{LL2}}, \mathbf{hf1_{LH1}}, \mathbf{hf1_{HL1}}, \mathbf{hf1_{HH1}}) \tag{19}$$
$$\mathbf{f_{rc2}} = \mathrm{IWT}(\mathbf{f_{LL1}}, \mathbf{hf1_{LH2}}, \mathbf{hf1_{HL2}}, \mathbf{hf1_{HH2}}) \tag{20}$$

where $\mathbf{f}_{\{c\}\{i\}}$ and $\mathbf{hf}_{\{c\}\{i\}}$ denote the frequency components c of number ith branch, e.g., $\mathbf{f_{LL2}}$ denotes the $\mathbf{f_{LL}}$ of T2 branch.

IV. Experimental Results

Experiments are mainly conducted on three datasets to evaluate the effectiveness of FWISN for CRCD. These include two generated simulated CRCD datasets, created by downsampling high-resolution RSI to produce low-resolution versions, and one real-world dataset captured by different satellite sensors. We compare our proposed method with 13 other methods, including ten traditional CD methods adapted for cross-resolution scenarios and three methods specifically designed for CRCD.

A. Datasets

1) LEVIR-CD (4×): LEVIR-CD consists of 637 RSI pairs with a size of 1024 × 1024 (0.5-m/pixel spatial resolution), which is one of the most widely used RSI datasets for CD. These image pairs are divided into 445/64/128 samples for train/validate/test [51]. In addition, each sample is cropped into patches with a size of 256 × 256. LEVIR-CD (4×) is the simulated CRCD dataset based on LEVIR-CD. The postevent (T2) image in each bitemporal RSI pair is downsampled by a factor of 4× to adapt cross-resolution scenarios. As a consequence, the simulated LEVIR-CD (4×) dataset consists of a series of image pairs with different resolution ratios, where the pre-event (T1) image has four times the spatial resolution of the postevent (T2) image.

2) SV-CD (8×): The season-varying CD (SV-CD) dataset [52] comprises 11 seasonally varying image pairs obtained from Google Earth. Of these, seven pairs have dimensions of 4725 × 2200 pixels, while the remaining four pairs measure 1900 × 1000 pixels. SV-CD (8×) simulated CRCD scenarios based on SV-CD. The T2 image in each bitemporal RSI pair is downsampled by a factor of 8×, while the resolution of the T1 image remains unchanged.

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3) DE-CD (3.3×): DE-CD (3.3×) [35] is collected from DynamicEarthNet [53], a real-world CRCD dataset. DynamicEarthNet provides periodic RSI of specific areas with varying resolutions (3 m/pixel for daily RSI and 10 m/pixel for monthly RSI). DE-CD contains 506 pairs of images sampled from DynamicEarthNet, with 3.3× resolution ratios. Each sample is cropped into nonoverlapping 256 × 256 patches from the original 1024 × 1024 image size. The DE-CD (3.3×) is default split into 354/51/101 samples for train/validate/test.

B. Experimental Setup

1) Implementation Details: We apply the same data augmentation techniques as described in [28] during training. We implement our proposed FWISN using the PyTorch framework. The training and evaluation are conducted on a single NVIDIA RTX 3090 GPU. The batch size is set to 16, and the maximum number of epochs is 250. We employ the loss function proposed in [21], which jointly optimizes the network performance on CD and segmentation tasks. The AdamW optimizer is used with an initial learning rate of 1×10^{-3}, which is reduced by 0.5 if the $F1$-score does not improve for ten consecutive epochs. The checkpoint achieving the highest $F1$-score is selected for testing, with inference beginning after 50 epochs. Parameters are initialized using PyTorch's default settings, and the random seed is set to 42 to ensure consistency.

2) Evaluation Metrics: To assess the performance of CD and CRCD, we adopt widely used evaluation metrics, including overall accuracy (OA), precision (P), recall (R), $F1$-score, and intersection over union (IoU). These metrics are standard for evaluating the effectiveness of CD models and other deep learning applications.

C. Overall Comparsion

We evaluate the proposed method and compare it with several advanced CRCD methods, including 11 traditional CD methods adapted to CRCD tasks and three specialized CRCD methods. These methods are evaluated on the LEVIR-CD (4×), SV-CD (8×), and DE-CD (3.3×) datasets. All these CD models are implemented using publicly available code and default hyperparameters. To adapt traditional CD methods for CRCD tasks, we resize low-resolution images to match high-resolution images using bicubic interpolation before training.

The performance degradation of conventional CD methods is reported in Table I. The methods in Table I all apply the Siamese network structure except FC-EF [54]. Performance degradation can be observed in Siamese networks when adapted to the CRCD task. In contrast, FE-EF shows limited performance degradation when adapted to the CRCD task. Although our method also exhibits performance degradation, it shows notable improvement compared with Siamese and non-Siamese networks. This results from exploiting the unequal relationship between time and space to extract resolution-invariant features.

The overall comparison is presented in Table II. The quantitative results demonstrate that our proposed method achieves higher scores and consistently outperforms conventional CD and CRCD methods across all three datasets. Specifically, our method achieves $F1$-score/IoU of 90.53/82.69, 96.82/93.83, and 50.62/33.96 on the LEVIR-CD, SV-CD, and DE-CD datasets, respectively.

TABLE I
PERFORMANCE DEGRADATION ON THE LEVIR-CD TEST SET. WE SELECT LEVIR-CD AND LEVIR-CD (4×) AS THE CONVENTIONAL CD AND CRCD DATASET. $F1$/IOU IS REPORTED, AND THE DIFFERENCE VALUE Δ IS SET AS THE METRIC OF PERFORMANCE DEGRADATION. FONT BOLD REPRESENTS THE BEST RESULT OF SIAMESE NETWORK STRUCTURE

	Conventional CD	**CRCD**	Δ
FC-EF [53]	75.79 / 61.01	75.31 / 60.4	0.48 / 0.61
FC-Siam-Conc [53]	82.28 / 69.9	76.4 / 61.81	5.88 / 8.09
FC-Siam-Diff [53]	79.17 / 65.52	70.83 / 54.83	8.34 / 10.69
SNUNet [54]	89.37 / 80.78	85.11 / 74.09	4.26 / 6.69
DMINet [55]	89.56 / 81.09	86.89 / 76.82	2.67 / 4.27
BiFA[56]	90.69 / 82.96	88.13 / 79.39	2.56 / 3.57
SRC-Net[59]	92.24 / 85.60	90.01 / 81.83	2.23 / 3.77
FWISN	**92.43 / 85.93**	**90.53 / 82.69**	**1.90 / 3.24**

Furthermore, we verify the adaptation ability of our model for continuous cross-resolution scenarios following [35]. The resolution ratio r_{d0} is used during training, but more resolution ratios are used during testing. For LEIVR-CD, SV-CD, and DE-CD, r_{d0} is set to 4, 8, and 3.3, and the continuous resolution ratios used for testing range from 1 to 8, 1 to 12, and 1 to 5, respectively. The ratio between 1 and r_{d0} is defined as the in-distribution ratio, while those exceeding r_{d0} are considered as out-of-distribution ratios. Tables III–V present the performance of various CD models on the LEVIR-CD, SV-CD, and DE-CD datasets. The quantitative results indicate that our proposed method consistently achieves optimal performance under continuous cross-resolution scenarios.

It can be observed that, as with other methods, our proposed method displays the arch distribution of metrics on continuous resolution tests, and the majority of methods demonstrate superior performance on in-distribution data compared with out-of-distribution data. However, our proposed method demonstrates a distinct advantage in terms of robustness under continuous cross-resolution scenarios. This is primarily evidenced by the following two points.

1) Nearly consistent performance is exhibited for each different resolution ratio setting. For example, resolution ratios from 1× to 5× on LEVIR-CD, 1× to 10× on SV-CD, and 1× to 3.3× on DE-CD.
2) For some extreme resolution cases, our method demonstrates superior performance. For example, 8.93/11.24 point increase in $F1$-score/IoU on the LEIVR-CD dataset with 8× resolution ratio and 3.38/6.07 and 4.37/7.68 point increase in $F1$-score/IoU on the SV-CD dataset with 10× and 12× resolution ratio.

This can be attributed to the fact that our method establishes unequal relationships in both the temporal and spatial domains. Regarding the temporal inequality, our approach focuses on images within a specific time domain, incorporating their features into separate branches. This strategy helps mitigate

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TABLE II

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON LEVIR-CD (4×), SV-CD (8×), AND DE-CD (3.3×) DATASETS. FONT RED, BLUE, AND BOLD REPRESENT THE OPTIMAL, SECOND-OPTIMAL, AND THIRD-OPTIMAL RESULTS, RESPECTIVELY

	LEVIR-CD(4×) Pre. / Rec. / F1 / IoU / OA	**SV-CD(8×)** Pre. / Rec. / F1 / IoU / OA	**DE-CD(3.3×)** Pre. / Rec. / F1 / IoU / OA
FC-EF [53]	79.57 / 71.48 / 75.31 / 60.40 / 97.64	74.25 / 45.03 / 56.06 / 38.95 / 91.67	-
FC-Siam-Conc [53]	84.23 / 69.90 / 76.40 / 61.81 / 97.82	73.11 / 50.87 / 59.99 / 42.85 / 92.00	-
FC-Siam-Diff [53]	86.12 / 60.15 / 70.83 / 54.83 / 97.50	76.10 / 56.68 / 64.97 / 48.12 / 92.79	-
STANet [50]	57.87 / 45.47 / 50.93 / 34.16 / 95.58	83.06 / 70.73 / 76.41 / 61.82 / 94.85	11.80 / **46.89** / 18.85 / 10.41 / 97.61
IFNet [57]	86.81 / 80.85 / 83.73 / 72.01 / 98.41	94.94 / 79.62 / 86.61 / 76.38 / 97.09	26.20 / 52.64 / 34.98 / 21.20 / 98.84
SNUNet [54]	89.67 / 81.00 / 85.11 / 74.09 / 98.57	92.61 / 83.80 / 87.98 / 78.55 / 97.30	38.21 / 37.16 / 37.68 / 23.21 / 99.27
BIT [28]	89.57 / 82.11 / 85.68 / 74.94 / 98.61	97.09 / 84.80 / 90.53 / 82.69 / 97.91	**62.05** / 33.38 / 43.41 / 27.72 / 99.48
ICIFNet [58]	87.84 / 84.62 / 86.20 / 75.75 / 98.63	95.68 / 90.56 / 93.05 / 87.00 / 98.40	**63.50** / 25.04 / 35.92 / 21.89 / 99.47
DMINet [55]	89.66 / 84.28 / 86.89 / 76.82 / 98.72	**97.77** / 89.76 / **93.60** / **87.96** / **98.55**	**71.47** / 33.84 / **45.93** / **29.81** / **99.53**
BiFA[56]	88.38 / 87.87 / 88.13 / **79.39** / **98.90**	- / - / - / - / -	- / - / - / - / -
SRC-Net[59]	**91.27** / **88.77** / **90.01** / **81.83** / **98.99**	- / - / - / - / -	- / - / - / - / -
SUNet [7]	64.12 / **93.54** / 76.08 / 61.40 / 97.03	63.55 / **97.98** / 77.10 / 62.73 / 93.13	32.60 / **71.00** / 44.68 / 28.77 / 98.96
SRCDNet [5]	66.29 / 84.18 / 74.17 / 58.94 / 97.04	91.30 / 91.89 / 91.59 / 84.49 / 98.01	39.62 / 33.22 / 36.13 / 22.05 / 99.30
SILI-CD [34]	**90.55** / 86.30 / **88.38** / 79.18 / 98.86	**95.29** / **93.36** / **94.32** / **89.24** / **98.67**	61.35 / 42.32 / **50.10** / **33.42** / **99.50**
Ours	**91.08** / **89.99** / **90.53** / **82.69** / **99.04**	**97.38** / **96.26** / **96.82** / **93.83** / **99.18**	51.83 / **52.20** / **52.02** / **35.15** / **99.43**

TABLE III

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON LEVIR-CD UNDER CONTINUOUS CROSS RESOLUTION. THESE MODELS ARE TRAINED ON LEVIR-CD (4×) AND TEST ON DIFFERENT RESOLUTION RATIOS. FONT RED, BLUE, AND BOLD REPRESENT THE OPTIMAL, SECOND-OPTIMAL, AND THIRD-OPTIMAL RESULTS, RESPECTIVELY

	In-distribution testing (F1 / IoU)					Out-of-distribution testing (F1 / IoU)		
	1×	**1.3×**	**2×**	**3×**	**4×**	**5×**	**6×**	**8×**
FC-EF [53]	73.67 / 58.31	74.49 / 59.35	75.11 / 60.14	75.49 / 60.62	75.31 / 60.40	74.41 / 59.24	72.52 / 56.88	66.83 / 50.18
FC-Siam-Conc [53]	78.63 / 64.78	79.33 / 65.74	79.32 / 65.73	78.34 / 64.39	76.40 / 61.81	73.55 / 58.16	69.91 / 53.73	61.57 / 44.48
FC-Siam-Diff [53]	75.39 / 60.51	76.29 / 61.66	76.06 / 61.36	74.17 / 58.94	70.83 / 54.83	66.12 / 49.39	60.35 / 43.22	46.57 / 30.35
STANet [50]	42.03 / 26.61	47.96 / 31.54	55.75 / 38.64	57.77 / 40.62	50.93 / 34.16	36.49 / 22.32	17.22 / 9.42	3.97 / 2.02
IFNet [57]	74.24 / 59.03	77.77 / 63.62	81.02 / 68.10	83.48 / 71.65	83.73 / 72.01	82.10 / 69.64	78.39 / 64.47	66.24 / 49.52
SNUNet [54]	85.13 / 74.11	86.77 / 76.62	87.52 / 77.81	87.24 / 77.36	85.11 / 74.09	76.63 / 62.12	59.56 / 42.41	17.66 / 9.68
BIT [28]	86.28 / 75.86	86.42 / 76.09	86.66 / 76.47	86.67 / 76.48	85.68 / 74.94	81.06 / 68.16	70.40 / 54.33	28.73 / 16.78
ICIFNet [58]	86.24 / 75.80	86.44 / 76.11	86.78 / 76.65	86.84 / 76.75	86.20 / 75.75	83.63 / 71.87	78.95 / 65.22	59.26 / 42.10
DMINet [55]	86.28 / 75.87	86.49 / 76.20	86.85 / 76.75	86.96 / 76.93	86.89 / 76.82	83.78 / 72.08	79.10 / 65.43	57.40 / 40.26
BiFA [56]	**87.68** / **78.66**	**87.93** / **79.07**	**88.21** / **79.52**	88.34 / **79.74**	88.13 / **79.39**	**86.80** / **77.25**	**84.53** / **73.72**	67.81 / 52.45
SRC-Net [59]	**87.34** / **77.53**	**88.22** / **78.92**	**89.46** / **80.93**	**90.06** / **81.92**	**90.01** / **81.83**	**89.18** / **80.47**	**86.64** / **77.00**	**80.15** / **67.72**
SUNet [7]	75.51 / 60.65	75.53 / 60.69	75.67 / 60.86	75.98 / 61.27	76.08 / 61.40	75.70 / 60.90	75.01 / 60.02	69.96 / 53.80
SRCDNet [5]	75.87 / 61.12	76.46 / 61.89	76.77 / 62.30	76.30 / 61.68	74.17 / 58.94	69.38 / 53.12	60.13 / 42.99	29.23 / 17.12
SILI-CD [34]	87.01 / 77.01	87.65 / 78.02	**88.21** / 78.90	**88.55** / 79.44	**88.38** / 79.18	86.73 / 76.57	84.31 / 72.87	**73.13** / **57.64**
Ours	**90.31** / **82.29**	**90.42** / **82.54**	**90.55** / **82.70**	**90.71** / **82.94**	**90.53** / **82.69**	**89.55** / **81.00**	**87.64** / **78.04**	**81.52** / **68.88**

TABLE IV

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON SV-CD UNDER CONTINUOUS CROSS RESOLUTION. THESE MODELS ARE TRAINED ON SV-CD (8×) AND TEST ON DIFFERENT RESOLUTION RATIOS. FONT RED, BLUE, AND BOLD REPRESENT THE OPTIMAL, SECOND-OPTIMAL, AND THIRD-OPTIMAL RESULTS, RESPECTIVELY

	In-distribution testing (F1 / IoU)					Out-of-distribution testing (F1 / IoU)		
	1×	**2×**	**4×**	**5×**	**8×**	**9×**	**10×**	**12×**
FC-EF [53]	55.88 / 38.77	55.88 / 38.77	55.96 / 38.85	55.99 / 38.88	56.06 / 38.95	56.06 / 38.94	56.06 / 38.95	56.03 / 38.92
FC-Siam-Conc [53]	64.31 / 47.39	64.52 / 47.62	63.97 / 47.03	63.35 / 46.36	59.99 / 42.85	58.39 / 41.23	56.84 / 39.70	53.88 / 36.87
FC-Siam-Diff [53]	67.35 / 50.77	67.41 / 50.84	67.09 / 50.48	66.86 / 50.22	64.97 / 48.12	63.54 / 46.56	61.84 / 44.76	58.51 / 41.35
STANet [50]	72.22 / 56.52	73.39 / 57.97	76.97 / 62.56	77.72 / 63.55	76.41 / 61.82	73.93 / 58.64	71.35 / 55.47	67.05 / 50.43
IFNet [57]	81.54 / 68.83	81.97 / 69.45	84.17 / 72.66	85.34 / 74.42	86.61 / 76.38	86.34 / 75.97	85.69 / 74.97	83.77 / 72.07
SNUNet [54]	75.08 / 60.10	79.29 / 65.69	87.77 / 78.21	89.62 / 81.20	87.98 / 78.55	85.54 / 74.74	83.45 / 71.60	80.04 / 66.73
BIT [28]	85.59 / 74.81	86.82 / 76.70	90.07 / 81.94	90.98 / 83.46	90.53 / 82.69	88.11 / 78.75	85.16 / 74.15	80.07 / 66.77
ICIFNet [58]	91.20 / 83.83	91.56 / 84.44	92.83 / 86.63	93.25 / 87.35	93.05 / 87.00	91.95 / 85.10	90.65 / 82.90	88.02 / 78.60
DMINet [55]	**92.14** / **85.42**	**92.52** / **86.09**	**93.66** / **88.07**	**93.92** / **88.54**	**93.60** / **87.96**	**93.00** / **86.91**	**92.05** / **85.27**	**89.59** / **81.14**
SUNet [7]	67.12 / 50.51	67.33 / 50.76	69.88 / 53.70	72.44 / 56.80	77.10 / 62.73	77.55 / 63.33	77.69 / 63.52	76.90 / 62.46
SRCDNet [5]	78.14 / 64.13	82.07 / 69.59	89.19 / 80.49	90.67 / 82.93	91.59 / 84.49	90.76 / 83.08	89.29 / 80.66	85.19 / 74.19
SILI-CD [34]	**94.07** / **88.80**	**94.11** / **88.88**	**94.26** / **89.14**	**94.30** / **89.22**	**94.32** / **89.24**	**93.55** / **87.87**	**92.80** / **86.57**	**90.50** / **82.65**
Ours	**94.25** / **89.14**	**94.72** / **89.97**	**96.11** / **92.50**	**96.53** / **93.30**	**96.82** / **93.83**	**96.58** / **93.39**	**96.18** / **92.64**	**94.87** / **90.24**

the impact of changes in data distribution caused by resolution differences. As for the spatial inequality, our method emphasizes high-frequency information, which is less sensitive to resolution variations, thereby enhancing the model's robustness.

The visualization results of several CRCD methods with varying bitemporal resolution difference ratios in CRCD datasets are presented in Fig. 6. This comparison provides additional evidence of the adaptability of our proposed method to cross-resolution scenarios. For example, in the first set of images, our method demonstrates superior performance by accurately detecting a cluster of buildings at all resolutions, whereas other methods fail to identify some of the structures effectively. Similarly, in the second set of images, our proposed method precisely predicts the elongated building in the lower left corner, while other methods exhibit false negatives (FNs).

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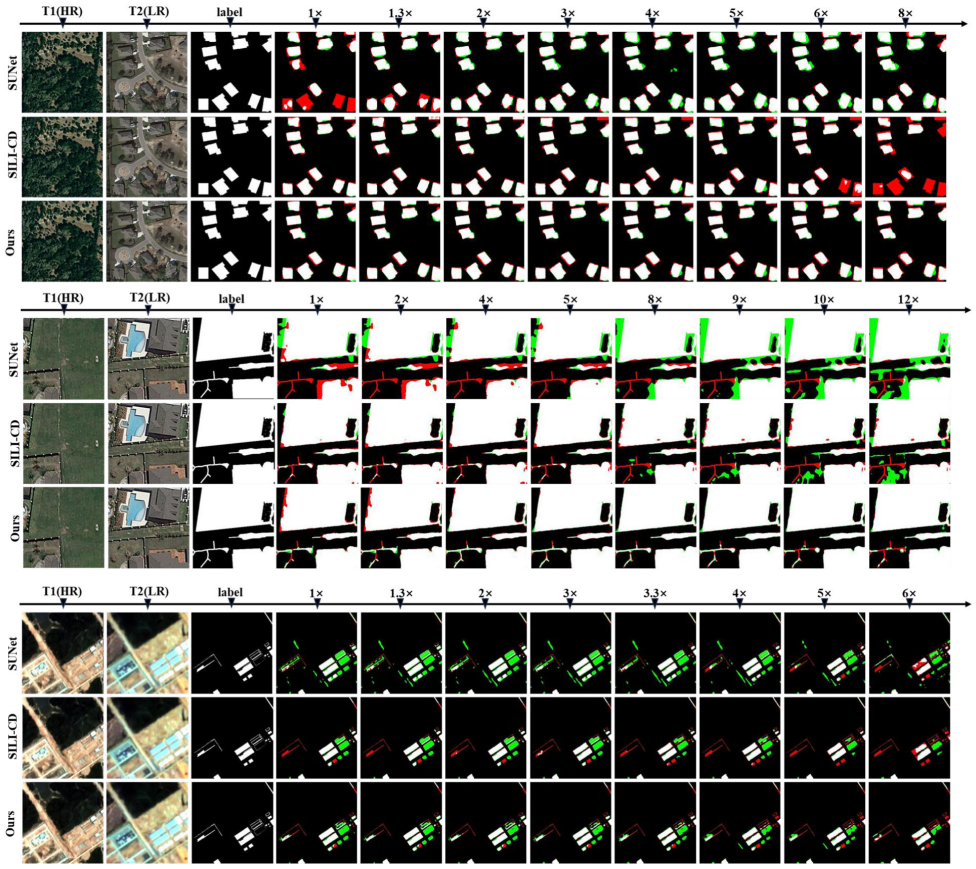


Fig. 6. We present a visual comparison of three methods on the three CD datasets with different resolution ratios. (Top to bottom) Visualization results are for LIVIR-CD, SV-CD, and DE-CD. For clearer visualization, true positives (TPs) are marked in white, true negatives are marked in black, false positives (FPs) are marked in red, and FNs are marked in green.

TABLE V

PERFORMANCE COMPARISON OF DIFFERENT METHODS ON DE-CD UNDER CONTINUOUS CROSS RESOLUTION. THESE MODELS ARE TRAINED ON DE-CD (3.3×) AND TEST ON DIFFERENT RESOLUTION RATIOS. FONT RED, BLUE, AND BOLD REPRESENT THE OPTIMAL, SECOND-OPTIMAL, AND THIRD-OPTIMAL RESULTS, RESPECTIVELY

	In-distribution testing (F1 / IoU)					Out-of-distribution testing (F1 / IoU)		
	1×	**1.3×**	**2×**	**3×**	**3.3×**	**4×**	**5×**	**6×**
STANet [50]	18.81 / 10.38	18.72 / 10.33	18.78 / 10.37	18.82 / 10.39	18.85 / 10.41	19.05 / 10.53	18.48 / 10.18	18.18 / 10.00
IFNet [57]	32.35 / 19.30	34.18 / 20.62	34.66 / 20.96	34.84 / 21.09	34.98 / 21.20	29.05 / 17.00	23.77 / 13.49	20.21 / 11.24
SNUNet [54]	32.49 / 19.40	35.76 / 21.77	37.43 / 23.02	37.69 / 23.22	37.68 / 23.21	27.88 / 16.20	22.68 / 12.79	18.45 / 10.16
BIT [28]	39.53 / 24.63	42.24 / 26.78	43.18 / 27.53	43.35 / 27.68	43.41 / 27.72	38.94 / 24.18	34.30 / 20.70	30.20 / 17.78
ICIFNet [58]	31.84 / 18.93	34.79 / 21.06	35.80 / 21.80	35.91 / 21.89	35.92 / 21.89	32.21 / 19.20	30.15 / 17.75	29.31 / 17.17
DMINet [55]	31.60 / 18.77	30.89 / 18.27	30.26 / 17.83	30.09 / 17.71	30.02 / 17.66	29.31 / 17.17	27.30 / 15.81	23.74 / 13.47
SUNet [7]	**43.69 / 27.95**	**44.48 / 28.60**	**44.66 / 28.75**	**44.68 / 28.77**	**44.68 / 28.77**	**42.84 / 27.25**	**40.56 / 25.44**	**37.58 / 23.14**
SRCDNet [5]	34.44 / 20.80	35.46 / 21.55	35.96 / 21.92	36.10 / 22.02	36.13 / 22.05	32.14 / 19.14	29.35 / 17.20	28.28 / 16.47
SILI-CD [34]	**47.88 / 31.48**	**49.86 / 33.21**	**50.23 / 33.54**	**50.15 / 33.54**	**50.09 / 33.42**	**48.45 / 31.97**	**45.45 / 29.41**	**41.71 / 26.35**
Ours	**51.28 / 34.47**	**51.39 / 34.58**	**51.64 / 35.80**	**52.08 / 35.21**	**52.02 / 35.15**	**51.15 / 34.36**	**47.60 / 31.23**	**42.72 / 27.16**

TABLE VI

ABLATION STUDY OF EACH MODULE IN FWISN ON THREE CD DATASETS. $F1$-SCORE IS REPORTED

Model	HFA	W-E	W-D	LEVIR(4×)	SV(8×)	DE(3.3×)
Base	×	×	×	89.40	95.56	46.06
FWISN	✓	×	×	90.26	96.37	48.24
FWISN	✓	✓	×	90.45	96.59	49.15
FWISN	✓	✓	✓	**90.53**	**96.82**	**52.02**

These findings collectively highlight the advanced performance of our approach, particularly for continuous CRCD tasks.

D. Ablation Study and Parametric Study

This section presents ablation and parametric experiments. First, we incrementally integrate various modules into the FWISN to assess and compare its performance, thereby validating the positive impact of the proposed strategies. Specifically, to evaluate the effectiveness of each module, HFA, WFB-E, and WFB-D are integrated incrementally to compare their respective performances. Second, we validate the impact of the spatial and temporal unequal relationship. Then, we conduct ablation studies on the application location of FFT and WT. Finally, we performed experiments that focused on the critical parameter of the model, i.e., the HF filtering threshold in HFA.

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TABLE VII
ABLATION STUDY OF OUR FWISN ON LEVIR-CD (4×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS

Model	HFA	WFB-E	WFB-D	1×	1.3×	2×	3×	4×	5×	6×	8×
Base	×	×	×	87.38	88.60	89.31	89.59	89.40	88.36	86.64	78.94
FWIS	✓	×	×	89.79	89.96	90.19	90.35	90.26	89.38	87.50	76.64
FWIS	✓	✓	×	90.08	90.24	90.44	90.58	90.46	**89.57**	**87.88**	**82.41**
FWIS	✓	✓	✓	**90.31**	**90.42**	**90.55**	**90.71**	**90.53**	89.55	87.64	81.52

TABLE VIII
ABLATION STUDY OF OUR FWISN ON SV-CD (8×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS

Model	HFA	WFB-E	WFB-D	1×	2×	4×	5×	8×	9×	10×	12×
Base	×	×	×	91.27	91.78	94.09	94.98	95.56	95.04	94.03	91.05
FWIS	✓	×	×	94.05	94.31	95.37	96.03	96.37	96.08	95.55	94.01
FWIS	✓	✓	×	94.04	94.13	95.75	96.25	96.59	96.32	95.84	**94.24**
FWIS	✓	✓	✓	**94.25**	**94.72**	**96.11**	**96.53**	**96.82**	**96.58**	**96.18**	**94.24**

TABLE IX
ABLATION STUDY OF OUR FWISN ON DE-CD (3.3×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS

Model	HFA	WFB-E	WFB-D	1×	1.3×	2×	3×	3.3×	4×	5×	6×
Base	×	×	×	45.09	45.26	45.60	46.25	46.06	44.92	41.71	36.34
FWIS	✓	×	×	47.07	47.21	47.56	48.12	48.24	47.17	44.14	40.00
FWIS	✓	✓	×	48.26	48.35	48.61	49.11	49.15	48.25	45.10	41.18
FWIS	✓	✓	✓	**51.28**	**51.39**	**51,64**	**52.08**	**52.02**	**51.15**	**47.60**	**42.72**

1) Effects of Individual Modules: To validate the effectiveness of each individual module, we incrementally integrate each of the three modules and evaluate their performance on the LEVIR-CD (4×) dataset. Table VI shows the ablation results of the components of our proposed method. These components essentially apply frequency-domain learning methods at different stages of the CD network: before the network, in the encoding stage, and in the feature fusion stage. Compared with baseline, frequency-domain learning methods bring significant improvements to the three datasets. In addition, we also conduct similar ablation experiments to evaluate the performance of our proposed method under continuous cross-resolution scenarios. Tables VII–IX show the results of the ablation experiments for FRC, WFB-E, and WFB-D under these scenarios.

Experimental results demonstrate that our method yields significant performance improvements on both the SV-CD and DE-CD datasets. On the LEVIR-CD dataset, while the HFA module enhances performance on in-distribution, we observe a decline in accuracy on out-of-distribution samples. A similar trend is seen with the WFB-D variant, which also exhibits reduced generalization when evaluated on out-of-distribution data.

We attribute this phenomenon to the intrinsic characteristics of the LEVIR-CD dataset. Specifically, most annotated changes are building-related, resulting in a limited diversity of change types. This narrow semantic scope may lead the model, particularly as its representational capacity increases, to overfit to the dominant patterns observed during training, thereby compromising its generalization to more diverse or structurally different changes.

TABLE X
ABLATION STUDY OF UNEQUAL RELATIONSHIPS ON LEVIR-CD (4×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS. BOLD FONTS AND UNDERSCORES INDICATE OPTIMAL AND SUBOPTIMAL RESULTS, RESPECTIVELY

	1×	1.3×	2×	3×	4×	5×	6×	8×
Base	87.38	88.60	89.31	89.59	89.40	88.36	86.64	78.94
Space-only	89.25	89.43	89.61	89.72	89.68	89.12	**88.13**	**80.04**
Time-only	**89.98**	**90.14**	**90.22**	**90.38**	**90.34**	86.53	83.41	68.62
HFA	89.79	89.96	90.19	90.35	90.26	**89.38**	87.50	76.64

TABLE XI
ABLATION STUDY OF UNEQUAL RELATIONSHIPS ON SV-CD (8×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS. BOLD FONTS AND UNDERSCORES INDICATE OPTIMAL AND SUBOPTIMAL RESULTS, RESPECTIVELY

	1×	2×	4×	5×	8×	9×	10×	12×
Base	91.27	91.78	94.09	94.98	95.56	95.04	94.03	91.05
Space-only	93.20	93.38	93.83	95.56	95.98	95.86	95.38	93.88
Time-only	93.41	93.87	95.19	95.64	95.98	95.73	95.30	93.91
HFA	**94.25**	**94.72**	**96.11**	**96.53**	**96.82**	**96.58**	**96.18**	**94.24**

Nevertheless, considering the consistent gains achieved on the other two datasets, and noting that the observed accuracy drop on LEVIR-CD out-of-distribution data remains within an acceptable range, we retain the WFB-D configuration and report the corresponding results for completeness.

2) Effects of Temporal and Spatial Unequal Relationship: We conduct an ablation study on unequal relationships in time and space. Tables X–XII report the performance of the unequal relationships in HFA. In space-only HFA, we remove the

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TABLE XII
ABLATION STUDY OF UNEQUAL RELATIONSHIPS ON DE-CD (3.3×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS. BOLD FONTS AND UNDERSCORES INDICATE OPTIMAL AND SUBOPTIMAL RESULTS, RESPECTIVELY

	1×	1.3×	2×	3×	3.3×	4×	5×	6×
Base	45.09	45.26	45.60	46.25	46.06	44.92	41.71	36.34
Space-only	46.31	46.41	46.71	47.22	47.29	46.36	43.42	39.85
Time-only	47.43	47.48	47.70	48.13	47.93	46.60	42.24	33.88
HFA	**51.28**	**51.39**	**51,64**	**52.08**	**52.02**	**51.15**	**47.60**	**42.72**

TABLE XIII
ABLATION STUDY OF FFT AND WT ON LEVIR-CD (4×), SV-CD (8×), AND DE-CD (3.3×) DATASETS. THE $F1$-SCORE IS REPORTED. BOLD FONTS INDICATE OPTIMAL RESULT. FRC AND CDNET REPRESENT FFT/WT FREQUENCY-DOMAIN MODELING METHOD IS USED AT THIS STAGE

FRC	CDNet	LEVIR-CD(4×)	SV-CD(8×)	DE-CD(3.3×)
FFT	FFT	90.02	95.56	46.34
WT	WT	90.13	96.17	47.03
FFT	WT	**90.53**	**96.82**	**52.02**

TABLE XIV
ABLATION STUDY OF FREQUENCY THRESHOLD ON LEVIR-CD (4×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS. BOLD FONTS AND UNDERSCORES INDICATE OPTIMAL AND SUBOPTIMAL RESULTS, RESPECTIVELY

	1×	1.3×	2×	3×	4×	5×	6×	8×
$R4$	89.88	90.01	90.21	90.43	90.48	89.48	87.54	79.28
$R8$	90.25	90.30	90.48	90.68	**90.60**	**89.59**	87.51	78.62
$R16$	**90.31**	**90.42**	**90.55**	**90.71**	90.53	89.55	**87.64**	**81.52**
$R32$	90.22	90.30	90.41	90.53	90.35	89.21	87.34	79.31

TABLE XV
ABLATION STUDY OF FREQUENCY THRESHOLD ON SV-CD (8×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS. BOLD FONTS AND UNDERSCORES INDICATE OPTIMAL AND SUBOPTIMAL RESULTS, RESPECTIVELY

	1×	2×	4×	5×	8×	9×	10×	12×
$R4$	93.77	94.25	95.56	95.92	96.10	95.81	95.34	93.82
$R8$	93.85	94.29	95.59	95.95	96.14	95.84	95.40	93.97
$R16$	**94.25**	**94.72**	**96.11**	**96.53**	**96.82**	**96.58**	**96.18**	**94.24**
$R32$	93.67	94.20	95.71	96.09	96.26	95.99	95.57	93.26

TABLE XVI
ABLATION STUDY OF FREQUENCY THRESHOLD ON DE-CD (3.3×) IN CONTINUOUS CROSS-RESOLUTION SCENARIOS. BOLD FONTS AND UNDERSCORES INDICATE OPTIMAL AND SUBOPTIMAL RESULTS, RESPECTIVELY

	1×	1.3×	2×	3×	3.3×	4×	5×	6×
$R4$	49.60	49.66	50.02	50.25	50.12	48.93	45.70	39.92
$R8$	49.65	49.72	49.92	50.56	50.67	50.65	47.24	42.71
$R16$	50.03	50.08	50.46	50.62	50.62	49.53	46.38	42.47
$R32$	**51.28**	**51.39**	**51,64**	**52.08**	**52.02**	**51.15**	**47.60**	**42.72**
$R64$	47.52	47.54	47.64	48.22	47.97	46.72	42.59	34.01

temporal unequal relationship, indicating that the reconstructed HF component comes from the same temporal domain. In Time-only HFA, we use the original images, rather than the HF components, for image reconstruction to eliminate spatially unequal relationships. Time-only HFA shows better performance at in-distribution resolutions but suffers significant degradation at out-of-distribution resolutions. Time-only HFA relies solely on information from a specific temporal domain (resolution), making it prone to overfitting for in-distribution resolutions. Space-only HFA exploits high-frequency information that is insensitive to resolution, thereby exhibiting advantages in out-of-distribution resolution ratio, but poorer performance in in-distribution resolution ratio. In contrast, HFA exploits unequal relationships in time and space and reports relatively balanced performance.

3) Effects of FFT and WT: We employ FFT in the image reconstruction stage and WT in the CD network. To assess the effects of these methods, we conduct an ablation study on FFT and WT in the FRC and CD network. We design three sets of experiments with FFT/FFT, WT/WT, and FFT/WT in FRC and CD network. The results are presented in Table XIII, which compares the performance of FFT and WT in the reconstruction stage and in the CD network. Quantitative results show that employing FFT in the FRC stage and WT in the CD network achieves the best performance. FFT is better at extracting global frequency-domain features than WT. These features are more significant in the FRC. In contrast, the CD network benefits from WT due to its emphasis on capturing local hierarchical features.

4) Analysis of the Frequency Threshold: In HFA, we set the frequency threshold of the high-pass filter to control the high-frequency component. In order to assess the effect of the threshold, we perform experiments on frequency thresholds. We adjust θ in (3). Tables XIV–XVI reports the result of different θ values. In these tables, different values of θ are denoted as $R\{\theta\}$ (e.g., $R4$ represents θ equal to 4). It can be observed that the performance of lower thresholds (e.g., $R4$ and $R8$) exhibits minimal difference compared with higher thresholds under low ratios (e.g., 1×, 1.3×, and 2×). This phenomenon could be attributed to the LR images in low-ratio scenarios having relatively abundant high-frequency details, where a lower threshold is already sufficient to capture the necessary features. However, as the resolution ratio increases (e.g., 9×, 10×, and 12×), high-frequency details in the LR images become increasingly scarce, and the preservation of high-frequency components becomes more critical. Lower thresholds fail to effectively retain these details, leading to degraded performance. In contrast, as the threshold increases, HFA retains more high-frequency information. When the threshold approaches infinity, the process degenerates into a Time-only HFA. As discussed above, HFA tends to overfit in-distribution data as the threshold grows, resulting in a decline in performance on out-of-distribution data. Therefore, in LEVIR-CD and SV-CD, the model tends to achieve optimal results at $R16$. However, in DE-CD, the model tends to achieve optimal results at $R32$, which may be due to the fact that DE-CD has a higher spatial resolution (3 m/pixel in DE-CD but 0.5 m/pixel in LEVIR-CD), and therefore requires a higher frequency threshold.

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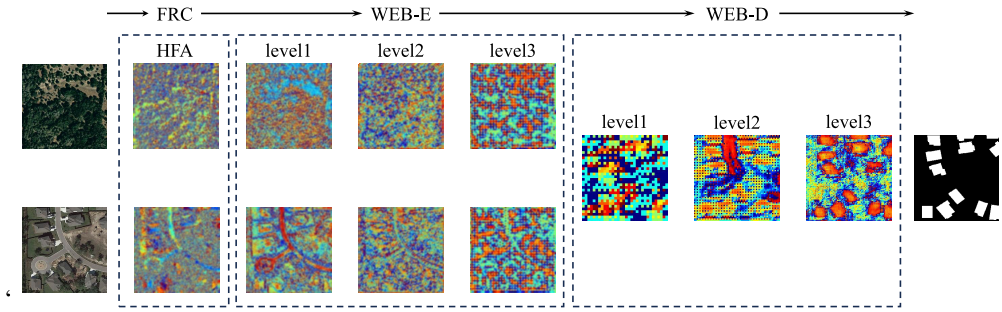


Fig. 7. Feature visualizations of our models are provided, with bitemporal feature differences displayed to show the effectiveness of our proposed methods.

E. Visualization

In this part, we present an example to illustrate the multilevel features derived from our model, demonstrating the effectiveness of our proposed methods. We visualize the attention maps from several stages of the FWISN to show the learning characteristics of the model on each encoder and decoder. These attention maps are channelwise weighted sums of activation maps from selected layers, focusing on the final layer of each encoder stage.

Fig. 7 illustrates the attention maps of our models. In this representation, red indicates elevated values, while blue corresponds to diminished values. The input sample is sourced from the LEVIR-CD (4×) test set. The attention map offers insights into the module's behavior, particularly following the HFA, where it reveals significant high-frequency information. Furthermore, it demonstrates a consistent high-frequency distribution between the images of T1 and T2, suggesting that T1 effectively identifies the spatial locations of high-frequency details present in T2.

V. Disscussion and Conclusion

In this article, we first explore the performance degradation in Siamese networks dealing with CD and conduct further analysis from a frequency-domain perspective. Our research identifies a novel resolution-invariant temporal and spatial unequal relationship from the frequency-domain perspective. Unlike existing methods that primarily focus on resolution alignment or feature space sharing, we highlight the critical role of high-frequency components in specific temporal domains, which has not been explicitly addressed in prior work. By introducing Fourier and WT mechanisms, our work bridges the gap between frequency-domain analysis and spatial-domain feature extraction. This integration provides a new perspective for addressing challenges in CRCD, offering a foundation for future research in remote sensing and beyond.

We propose FWISN, which incorporates an FRC stage, utilizing the HFA module to extract and reconstruct high-frequency features. In addition, we propose the WFB module and integrate it into both the encoder and decoder to enhance resolution-invariant features. The FWISN framework demonstrates state-of-the-art performance on three datasets, including a real-world CRCD dataset. The experimental results indicate that FWISN handles varying resolution ratios effectively, which makes it well suited for CRCD and real-world applications.

VI. Future Work

While FWISN has demonstrated promising results in handling resolution-invariant feature representation through frequency-based mechanisms, several avenues remain open for further exploration. One potential direction is the incorporation of adaptive frequency learning, where the network dynamically adjusts its sensitivity to frequency components based on input-specific or domain-specific priors. This could further enhance the model's robustness against diverse spatiotemporal distortions frequently encountered in real-world CRCD scenarios.

Another promising extension lies in the domain of semantic CD (SCD), where not only the occurrence of change but also the semantic nature or class of the change is of interest. Integrating the frequency-aware modules (e.g., HFA and WFB) with semantic decoders or transformer-based label propagation mechanisms may help disentangle subtle semantic shifts across different spatial resolutions and time intervals. Moreover, frequency-domain features could serve as complementary signals to boost category-level consistency and boundary delineation in SCD tasks.

Finally, a more theoretical understanding of the interaction between frequency components and high-level semantic structures remains an open question. Investigating the interpretability of frequency-aware modules could lead to more explainable models and novel frequency-based priors for remote sensing tasks, fostering a broader application in environmental monitoring, urban development tracking, and disaster assessment.

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